

STUDENT MANAGER APP — BACKEND DOMAIN MODEL

UML Class Diagram

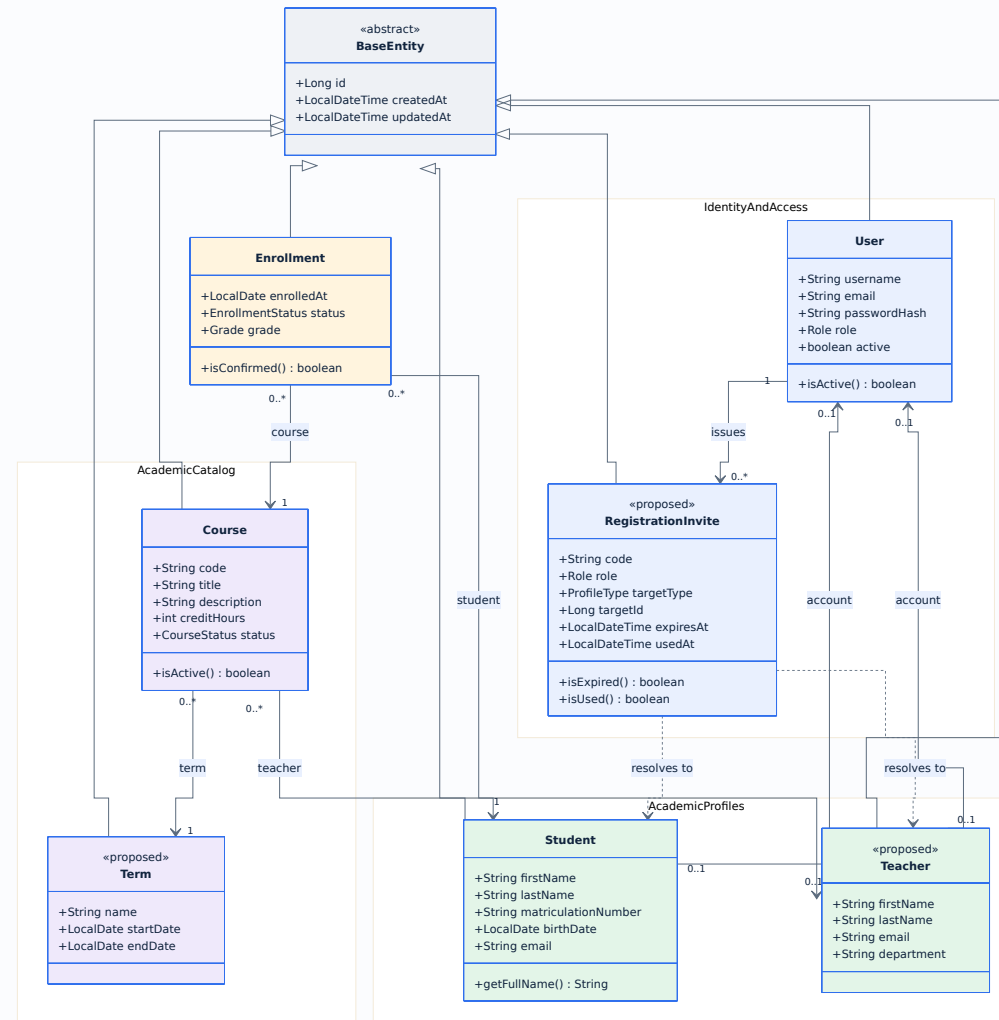
Object-level view of the domain: classes, attributes, key behaviour, and how instances relate to one another — inheritance from `BaseEntity`, optional 1:1 account links, and the enums each class depends on.

Green-labelled classes exist in the codebase today; amber-labelled classes (`Teacher`, `Term`, `RegistrationInvite`, `ProfileType`) are part of the proposed redesign, not yet implemented.

Domain Class Diagram

Solid arrow = navigable association with multiplicity (the class at the tail holds the reference). Hollow triangle = inheritance from `BaseEntity`. Dashed arrow = dependency — for `RegistrationInvite`, a reference resolved in application code rather than a real object link, since `targetType/targetId` is polymorphic. Enum-typed attributes (`Role`, `CourseStatus`, `EnrollmentStatus`, `Grade`, `ProfileType`) are shown inline rather than as separate boxes, to keep the relationship graph readable — their values are listed on the next page. A «proposed» tag marks classes not yet in the codebase.

 Base / audit  Identity & access  Academic profiles  Academic catalog  Association (Enrollment)



Legend & Class Notes

	Inheritance: subclass extends <code>BaseEntity</code> (id + audit timestamps).		Navigable association: the source class holds a field referencing the target.
"0..1"	Multiplicity: how many instances of the other class this end relates to.		(dashed) Dependency: uses the type, or resolves to it only in code (polymorphic).

Class	Status	Represents
User	existing	Login identity and role — no academic data.
Student	existing	Academic profile; gains an optional user link in the redesign.
Course	existing	A course offering; gains term and teacher in the redesign.
Enrollment	existing	Association class between Student and Course, carrying status + grade.
Teacher	proposed	New — gives the TEACHER role an actual profile and course ownership.
Term	proposed	New — scopes courses/enrollments to an academic period.
RegistrationInvite	proposed	New — the admin-issued, one-time code that links a fresh User to a pre-created Student/Teacher.
ProfileType	proposed	New enum — disambiguates what <code>RegistrationInvite.targetId</code> points to.

Enumeration	Values	Used by
Role	STUDENT, TEACHER, ADMIN	User.role, RegistrationInvite.role
ProfileType	STUDENT, TEACHER	RegistrationInvite.targetType
CourseStatus	ACTIVE, INACTIVE, ARCHIVED	Course.status
EnrollmentStatus	PENDING, CONFIRMED, CANCELLED	Enrollment.status
Grade	A, B, C, D, F, NOT_GRADED	Enrollment.grade